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# FLAVOURBENCH: What Language Models Know and What Epicure Adds

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## Abstract

1 Culinary benchmarks usually rely on static question banks or model judges.  
2 FLAVOURBENCH uses a versioned EPICURE runtime instead. Four determin-  
3 istic EPICURE operations produce 32 multiple-choice tasks and their answer keys  
4 before any evaluated model is called. No language model writes or judges the test.  
5 For this benchmark, ground truth means agreement with that pinned runtime.  
6 Each of 20 model endpoints receives every task twice. In the *Model only* condition,  
7 exact accuracy is the FlavourBench Score. In the *Model + Epicure* condition, the  
8 model may make one read-only call to the named operation. The difference is  
9 Epicure Gain. The release contains 640 matched pairs and 1280 scored responses.  
10 GPT-5.6 Sol Pro leads with a FlavourBench Score of 62.5. Across the panel,  
11 Epicure changes accuracy from 43.9% to 93.9%, a gain of +50.0 percentage points.  
12 The assisted condition measures whether a model can use a specified operation, not  
13 whether it can discover the right tool.  
14 A single content-addressed release contains the tasks, answer keys, routes, re-  
15 sponses, and bounded Epicure traces. It reconstructs every score, table, and figure  
16 offline. FLAVOURBENCH therefore separates an operational model score from the  
17 measured benefit of a domain system, without a human or model judge.

## 18 1 Introduction

19 Benchmarks are easiest to trust when their ground truth is external to the models being tested. Code  
20 can be executed, database states can be inspected, and games can be scored. Culinary reasoning  
21 rarely has that luxury. Ingredient substitutions, pairings, technique choices, and cuisine associations  
22 are commonly assessed by another language model or by a small set of human preferences. The  
23 evaluator then becomes part of the construct.

24 FLAVOURBENCH starts from EPICURE, a read-only culinary runtime with 1,790 ingredients repre-  
25 sented in a 300-dimensional space [1, 2]. Four runtime operations are converted into four-choice  
26 questions. The compiler records each answer before it contacts an evaluated model.

27 The design asks two separate questions. The FlavourBench Score measures how often a model  
28 endpoint returns EPICURE’s answer without access to the runtime. Epicure Gain measures the change  
29 when the same endpoint receives the same task with one read-only EPICURE call. Every model  
30 receives every task in both conditions. The score determines the leaderboard; Epicure Gain does not  
31 enter the ranking.

32 Our release evaluates 20 current endpoints, including GPT-5.6 Sol, Terra, and Luna; Claude Fable,  
33 Opus, and Sonnet 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.5; Kimi K3; Qwen 3.8 Max; GLM 5.2;  
34 two DeepSeek V4 variants; MiniMax M3; Nemotron 3 Ultra; Mistral Large 2512; Tencent HY 3;  
35 and Cohere Command A Plus and Command A Reasoning. Kimi and both Cohere models use direct

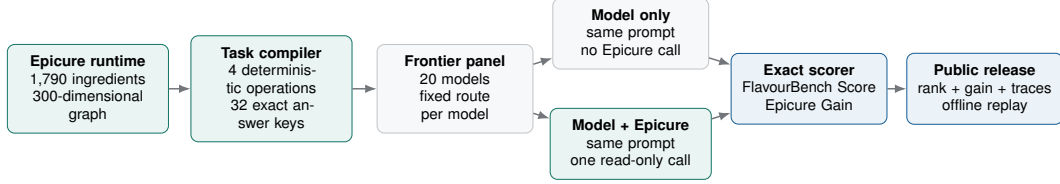


Figure 1: The FLAVOURBENCH pipeline. EPICURE fixes each answer before evaluation. The Model only arm produces the ranked FlavourBench Score. The matched Model + Epicure arm produces Epicure Gain. Both are reconstructed from the same public release.

| Family       | Epicure operation             | Exact target              | Example query                  | Tasks |
|--------------|-------------------------------|---------------------------|--------------------------------|-------|
| Substitution | <code>neighbors</code>        | first ranked neighbour    | top neighbour for tomato       | 8     |
| Composition  | <code>pairing_score</code>    | four-decimal affinity     | tomato + basil                 | 8     |
| Cookability  | <code>compare_on_axis</code>  | larger flavour projection | coffee vs. chocolate on bitter | 8     |
| Evidence     | <code>cultural_profile</code> | highest cuisine cosine    | cuisine direction for kimchi   | 8     |

Table 1: The task set is balanced by construction. Each family maps one deterministic EPICURE operation to a four-choice answer. Chance accuracy is 25%.

provider routes. Every other endpoint uses an exact, no-fallback OpenRouter route selected before the run.

The paper makes three contributions:

1. an automated culinary benchmark whose questions and answer keys come from a versioned executable system rather than a model judge;
2. a common-task frontier evaluation with 640 matched pairs, exact missing-response penalties, and a public 20-row leaderboard;
3. a matched Epicure intervention that reports panel gain and task-family interactions without changing the ranking score.

EPICURE has two roles. It fixes the answer key, then becomes the intervention in the second arm. This keeps the leaderboard and the Epicure comparison connected without combining them into an arbitrary composite score.

## 2 Epicure reference answers

### 2.1 A versioned reference

For FLAVOURBENCH, ground truth is the output of a specific version of EPICURE. It is not a claim about universal culinary preference. The release binds the application, evidence bundle, release identifier, ingredient count, and embedding dimensionality by hash. A different runtime produces a different task-set hash.

This definition makes every answer independently recomputable. It also gives the score a direct interpretation: accuracy is agreement with the published culinary runtime under the published task construction. No second model decides which response sounds better.

### 2.2 Four balanced task families

The compiler creates eight tasks in each of four families. Seeds and ingredient pairs are fixed before evaluation. For every task, EPICURE is called once to obtain the reference result. The correct answer is extracted programmatically, three distractors are constructed without model generation, and the four options are shuffled with a task-ID-derived seed.

The task families deliberately mix retrieval and computation. Neighbour and cuisine questions test whether a model knows relationships encoded by the graph. Pairing questions require an exact numeric result that is difficult to guess. Axis questions test a comparative claim rather than a

65 memorised scalar. Together they prevent the leaderboard from reducing to one operation or one kind  
66 of culinary trivia.

### 67 2.3 Prompt and answer contract

68 Each prompt names the relevant operation and gives four choices. Models may explain briefly, but  
69 the last answer line must be FINAL\_CHOICE: X. The parser accepts A, B, C, or D on that marker  
70 line. A missing marker, abnormal provider finish, timeout, or absent response scores zero. The rule  
71 fits in the standalone verifier shipped with the release.

72 The complete task set contains no hidden human label and no model-authored item. The benchmark is  
73 therefore suitable for automatic reruns, CI-style regression tests, and training loops where thousands  
74 of candidate responses must be scored consistently.

## 75 3 Evaluation design

### 76 3.1 One score and one matched intervention

77 For model  $m$  and task  $t$ , let  $M_{mt}$  equal one when the Model only response finishes normally and  
78 matches the EPICURE answer. Let  $E_{mt}$  be the same indicator for Model + Epicure. The prompt,  
79 route, temperature, top- $p$ , seed request, output limit, and answer parser are identical. Only access to  
80 EPICURE changes.

81 Each Model + Epicure response may make one read-only call to the operation named in the prompt.  
82 The runner records the operation, arguments, latency, error flag, and a hash of the returned payload.  
83 The returned data is bounded before it enters model context. Automatic provider fallback is disabled,  
84 so a route failure remains a failure for that route.

85 The single ranking metric is the FlavourBench Score:

$$F_m = 100 \frac{1}{32} \sum_t M_{mt}. \quad (1)$$

86 Every family contains eight tasks, so ordinary accuracy already gives each family equal weight.  
87 Missing or invalid responses remain in the denominator as incorrect. The public release sorts by  $F_m$ ,  
88 then uses Model + Epicure accuracy, Model + Epicure completion, and model ID as fixed tie-breakers.  
89 Epicure Gain is the matched difference

$$G_m = 100 \frac{1}{32} \sum_t (E_{mt} - M_{mt}), \quad (2)$$

90 reported in percentage points.  $G_m$  is not part of the FlavourBench Score. The release also keeps the  
91 four paired outcomes for every task: both correct, Model only correct, Model + Epicure correct, or  
92 neither correct.

### 93 3.2 Completion and uncertainty

94 Completion is the fraction of assigned responses that end normally with a parseable final marker.  
95 The score still counts missing responses as incorrect. Reporting completion beside the score keeps  
96 endpoint failure separate from a wrong culinary answer. A zero score with zero completions is not  
97 evidence of zero model capability. For each 32-task accuracy we report a 95% Wilson interval.

98 The benchmark is designed for visible differences, not hundredths of a point. One answer changes  
99 the score by 3.125 points. Family breakdowns and the full pair matrix are published so tied or nearby  
100 aggregate scores can be inspected rather than over-interpreted.

### 101 3.3 Twenty-model frontier panel

102 The panel was fixed before the final calls. It includes three GPT-5.6 tiers rather than treating one  
103 deployment as the entire OpenAI family; three Claude 5 tiers; two Gemini tiers; two DeepSeek  
104 V4 routes; both current Cohere Command A routes; and a broad set of current independent and

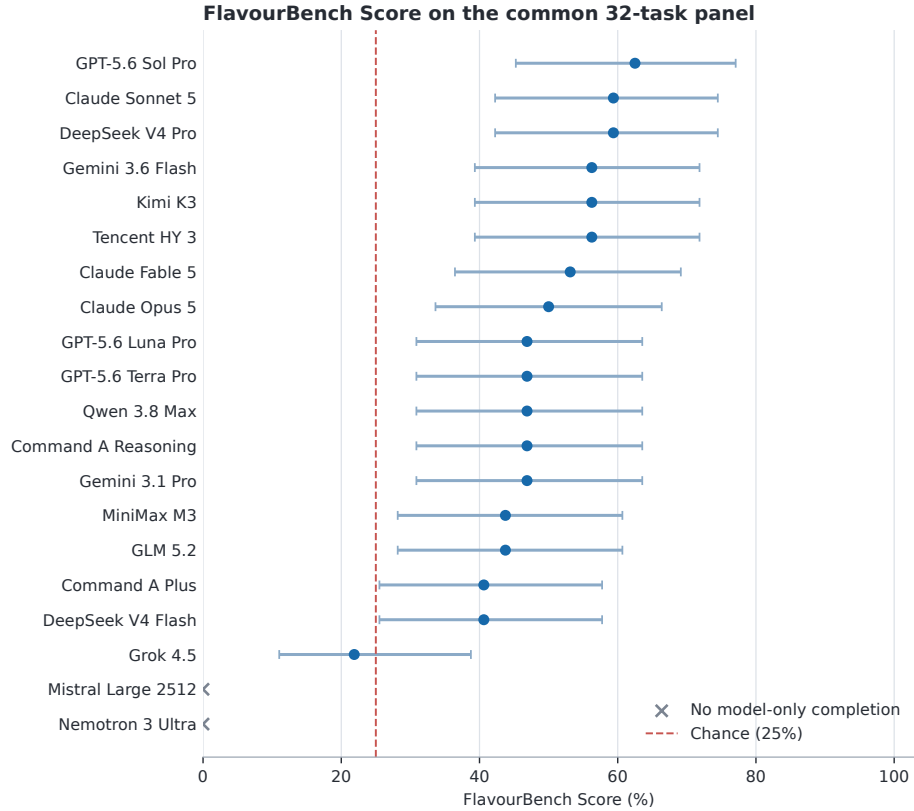


Figure 2: FlavourBench Score against EPICURE answer keys. Points are exact accuracy over the common 32-task panel; bars are 95% Wilson intervals. Crosses mark endpoints with no Model only completion. The dashed line is four-choice chance.

open-weight families. Kimi K3 runs through Kimi’s Anthropic-compatible endpoint. Command A Plus and Command A Reasoning run through Cohere V2 Chat. Qwen 3.8 Max and GLM 5.2 use explicit OpenRouter routes in the automated batch [5–7, 11, 14, 15]. The exact 20-row route table appears in Appendix C.

## 4 Results

### 4.1 The FlavourBench Score leaderboard

GPT-5.6 Sol Pro leads with 20/32 correct and a FlavourBench Score of 62.5. Table 2 gives the full release ordering and completion rates. Figure 2 adds intervals to show which gaps the 32-task panel can resolve. Models with no Model only completion are marked DNF rather than interpreted as the least capable systems.

The answer key exists before the model call, every endpoint receives the same questions, and no judge is sampled after the fact. Missing outputs count as incorrect in the operational score but remain visible in the completion columns. Mistral Large 2512 and Nemotron 3 Ultra returned no Model only completion in this run, so their zeroes are transport outcomes rather than capability estimates. Nemotron also returned no Model + Epicure completion.

### 4.2 What Epicure changes

Across all 640 matched pairs, panel accuracy is 43.9% for Model only and 93.9% for Model + Epicure. Epicure Gain is therefore +50.0 points. Every route that produced at least one assisted response gains points under the operational scoring rule. The largest numerical change is Mistral Large 2512 at

| Rank | Model               | FB Score | +Epicure | Gain  | Model % done | Epicure % done |
|------|---------------------|----------|----------|-------|--------------|----------------|
| 1    | GPT-5.6 Sol Pro     | 62.5     | 100.0    | +37.5 | 96.9         | 100.0          |
| 2    | Claude Sonnet 5     | 59.4     | 100.0    | +40.6 | 100.0        | 100.0          |
| 3    | DeepSeek V4 Pro     | 59.4     | 93.8     | +34.4 | 100.0        | 93.8           |
| 4    | Gemini 3.6 Flash    | 56.2     | 100.0    | +43.8 | 100.0        | 100.0          |
| 5    | Kimi K3             | 56.2     | 100.0    | +43.8 | 100.0        | 100.0          |
| 6    | Tencent HY 3        | 56.2     | 100.0    | +43.8 | 100.0        | 100.0          |
| 7    | Claude Fable 5      | 53.1     | 100.0    | +46.9 | 93.8         | 100.0          |
| 8    | Claude Opus 5       | 50.0     | 100.0    | +50.0 | 100.0        | 100.0          |
| 9    | GPT-5.6 Luna Pro    | 46.9     | 100.0    | +53.1 | 100.0        | 100.0          |
| 10   | GPT-5.6 Terra Pro   | 46.9     | 100.0    | +53.1 | 100.0        | 100.0          |
| 11   | Qwen 3.8 Max        | 46.9     | 100.0    | +53.1 | 87.5         | 100.0          |
| 12   | Command A Reasoning | 46.9     | 96.9     | +50.0 | 93.8         | 100.0          |
| 13   | Gemini 3.1 Pro      | 46.9     | 96.9     | +50.0 | 93.8         | 96.9           |
| 14   | MiniMax M3          | 43.8     | 100.0    | +56.2 | 96.9         | 100.0          |
| 15   | GLM 5.2             | 43.8     | 96.9     | +53.1 | 100.0        | 96.9           |
| 16   | Command A Plus      | 40.6     | 100.0    | +59.4 | 100.0        | 100.0          |
| 17   | DeepSeek V4 Flash   | 40.6     | 100.0    | +59.4 | 75.0         | 100.0          |
| 18   | Grok 4.5            | 21.9     | 100.0    | +78.1 | 31.2         | 100.0          |
| DNF  | Mistral Large 2512  | 0.0      | 93.8     | +93.8 | 0.0          | 100.0          |
| DNF  | Nemotron 3 Ultra    | 0.0      | 0.0      | +0.0  | 0.0          | 0.0            |

Table 2: The public leaderboard. “FB Score” is Model only exact-choice accuracy and determines rank. “+Epicure” is accuracy for the same tasks with EPICURE available. “Gain” is their difference in percentage points. The final columns report completion. DNF means that the Model only endpoint returned no parseable completion.

| Family       | Epicure operation       | Tasks | Model only | Model + Epicure |
|--------------|-------------------------|-------|------------|-----------------|
| Substitution | <b>neighbors</b>        | 8     | 18.1       | 94.4            |
| Composition  | <b>pairing_score</b>    | 8     | 40.6       | 94.4            |
| Cookability  | <b>compare_on_axis</b>  | 8     | 62.5       | 92.5            |
| Evidence     | <b>cultural_profile</b> | 8     | 54.4       | 94.4            |

Table 3: Panel-average performance by family and the corresponding reference operation.

+93.8 points, but that route had no Model only completion. Its change combines condition-specific availability with answer quality and should not be read as a pure knowledge gain.

The assisted prompt names the relevant operation. This is a scaffolded test of calling that operation, reading its output, and preserving the answer contract. It is not a test of open-ended tool discovery. That narrower intervention is useful here because it isolates the benefit of EPICURE from the separate problem of choosing among many tools.

### 4.3 Families explain aggregate ties

The four task families have different demands. Pairing questions require an exact four-decimal value; substitution and cultural questions can sometimes be answered from broad food knowledge; axis comparisons require a directional decision in EPICURE’s latent space. Figure 4 shows that similar total scores can arise from different profiles.

### 4.4 Every model and task pair is inspectable

Aggregate bars can hide a benchmark bug or a single pathological task. Figure 5 therefore publishes all 640 paired outcomes. Blue cells are stable knowledge: both conditions are correct. Gold cells are Epicure wins: Model only is wrong and Model + Epicure is correct. Red cells show the reverse outcome. Grey cells show questions neither condition solved.

### 4.5 Cohere, Kimi, Qwen, and route fidelity

Both Cohere models are full members of the public panel. Command A Plus ranks 16 with a FlavourBench Score of 40.6 and an Epicure Gain of +59.4 points. Command A Reasoning ranks 12

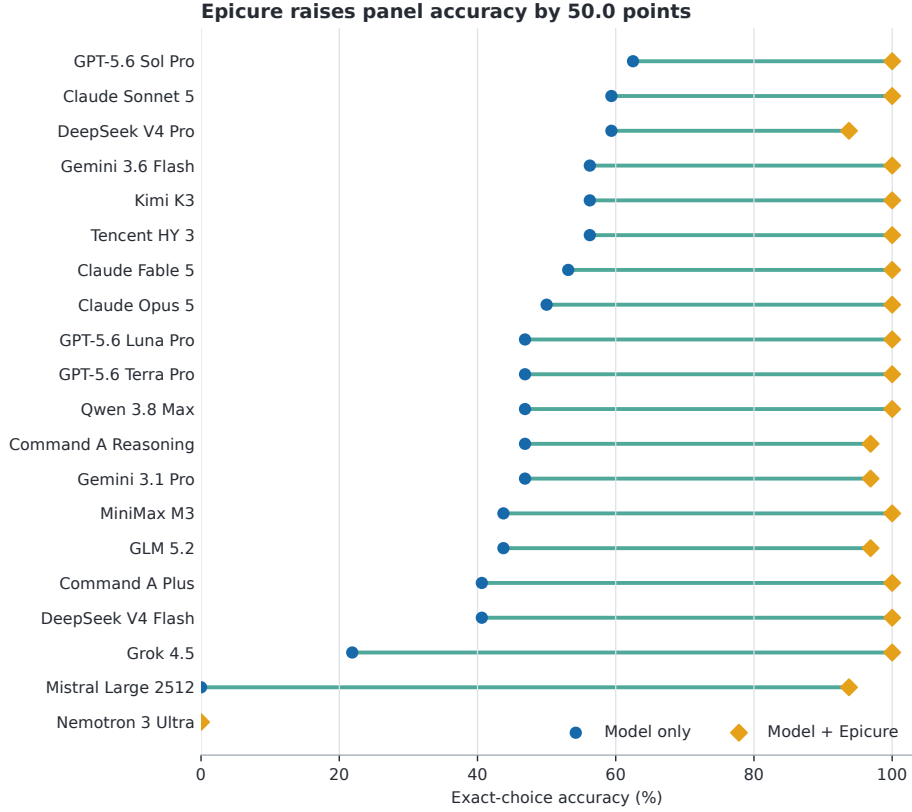


Figure 3: Model only and Model + Epicure accuracy over the same 32 tasks. Each line is a matched comparison within one endpoint, not a comparison between different task samples.

143 with a score of 46.9 and a gain of +50.0 points. Kimi K3 ranks 5 through the direct Kimi endpoint.  
 144 Qwen 3.8 Max ranks 11 through an exact OpenRouter route, and GLM 5.2 ranks 15. No route  
 145 changes during generation.

146 The two direct Cohere routes each produced all 64 assigned arms. Their returned identifiers are comm  
 147 and-a-plus-05-2026 and command-a-reasoning-08-2025, respectively; the direct Kimi route  
 148 likewise produced all 64 arms under returned identifier k3. The offline verifier checks these provider,  
 149 identity, and arm-count contracts before accepting the release.

150 This matters because a model name is not a transport contract. The release stores the requested model,  
 151 returned model, provider, route, generation ID, latency, token counts, and response hash. Direct and  
 152 routed endpoints can therefore be analysed separately without pretending they are interchangeable  
 153 deployments.

#### 154 4.6 Score, response time, and Epicure Gain

155 Figure 6 plots FlavourBench Score against median response time. Marker colour encodes Epicure  
 156 Gain. A fast model can score poorly, a high-scoring model can gain little from EPICURE, and a  
 157 lower-ranked model can improve sharply when assisted. The release keeps these dimensions separate.

### 158 5 Using the score as a training signal

159 The same environment exposes four binary indicators for a generated trajectory  $\tau$ :  $r_{\text{answer}}$  for a  
 160 correct answer,  $r_{\text{format}}$  for a parseable final marker,  $r_{\text{call}}$  for a successful Epicure call, and  $r_{\text{align}}$  for  
 161 a call matching the reference operation. The public leaderboard uses  $r_{\text{answer}}$  in Model only. A future

162 training objective could use

$$r(\tau) = r_{\text{answer}}r_{\text{format}} + \lambda r_{\text{answer}}r_{\text{call}} + \mu r_{\text{align}}, \quad (3)$$

163 where each component is a binary indicator and  $\lambda, \mu$  set the value of successful and reference-  
164 matching calls. Requiring a correct answer prevents irrelevant calls from earning the main reward.  
165 The interface could support rejection sampling, preference optimisation, or reinforcement learning on  
166 new ingredient seeds. This paper does not train a checkpoint; such a study should keep the public  
167 leaderboard as a final evaluation split.

## 168 6 Reproducibility and release

169 The public release is one JSON document with hash 197b84d75cca... It contains the 20 route  
170 records, 32 full prompts, answer keys and reference EPICURE results, 1,280 observation rows, final  
171 answer markers, response times, and bounded Epicure-trace projections. 1195 provider responses  
172 were observed; 85 missing arms remain explicit zero-scored rows.

173 The leaderboard can be reconstructed without a network connection, provider key, GPU, or running  
174 EPICURE service:

```
175 python3 -I reproduce_epicure_native.py \  
176     --release generated/epicure-native/  
177     epicure-native-release.json
```

178 The verifier rejects duplicate JSON keys, non-finite numbers, a changed content address, an incom-  
179 plete  $20 \times 32 \times 2$  grid, or any leaderboard value that differs from recomputation. The paper tables  
180 and figures are generated from the same release, so a changed score cannot leave a stale chart behind.

181 Reproducing the original hosted calls is a different operation from replaying the result. The release  
182 records exact routes and request policies, but provider deployments can change. A fresh frontier run  
183 should generate a new manifest and release rather than overwrite this snapshot.

## 184 7 Discussion

185 **What the score means.** The FlavourBench Score answers a narrow operational question: how  
186 often did this deployed endpoint return the answer fixed by the published EPICURE runtime? It  
187 includes endpoint availability because missing responses score zero. The completion columns reveal  
188 that component. The score is not a universal measure of cooking skill, but it is exact, reproducible,  
189 and independent of a model judge.

190 **Do we need humans?** Not for this leaderboard. Humans are useful when the target is preference,  
191 creativity, safety, or the quality of an open-ended recipe. They are unnecessary when the question  
192 is whether a model’s choice matches a deterministic runtime result. FLAVOURBENCH uses that  
193 distinction to automate the core ranking while leaving subjective culinary evaluation as a separate  
194 track.

195 **Why run Model + Epicure?** The leaderboard says which endpoint best predicts EPICURE. The  
196 matched intervention asks how much the same endpoint changes when given EPICURE directly.  
197 That result is operationally useful: a strong model may gain little, while a lower-scoring model may  
198 become competitive when grounded. The dumbbell and pair matrix show this task by task.

199 **Scaling the benchmark.** The task compiler can expand the seed lists while preserving the four  
200 operations. The next release should reserve unseen ingredients and pairs, remove near-duplicate  
201 questions, and measure stability across runtime versions. The current 32-task panel is a frontier  
202 snapshot and a reproducible template for a larger hidden set.

## 203 8 Related work

204 HELM argues for explicit scenarios and metrics, while recent benchmark audits emphasise validity,  
205 contamination, and transparent reporting [12, 19]. Chatbot Arena and WildBench evaluate broad

| Benchmark     | Scope              | Per-answer target           | Deterministic score | Matched Epicure intervention |
|---------------|--------------------|-----------------------------|---------------------|------------------------------|
| Chatbot Arena | general chat       | pairwise preference         | No                  | No                           |
| WildBench     | general chat       | task check-list             | No                  | No                           |
| BFCL          | function calling   | structured call             | Yes                 | No                           |
| $\tau$ -bench | interactive agents | exact goal state            | Yes                 | No                           |
| FLAVOURBENCH  | culinary reasoning | versioned EPICURE operation | Yes                 | Yes                          |

Table 4: FLAVOURBENCH complements broad preference and tool-use benchmarks. Its distinctive unit is the same endpoint and task measured as Model only and Model + Epicure.

answer quality through preferences or checklists [3, 13]. FLAVOURBENCH instead targets an executable domain model and needs no judge for its primary score.

ToolLLM, BFCL, and  $\tau$ -bench make tool invocation or environment state observable [16, 17, 24]. FLAVOURBENCH adds a matched Model only condition, so EPICURE becomes an intervention rather than a requirement shared by every response. LiveBench, LiveCodeBench, GPQA, MMLU-Pro, and SWE-bench illustrate complementary strategies for reducing contamination or grounding answers in difficult tasks and executable outcomes [9, 10, 18, 21, 22].

Culinary datasets such as FlavorDB and Recipe1M organise ingredient and recipe evidence [8, 20]. CookingSense and planning-oriented culinary work test commonsense and procedural reasoning [4, 23]. EPICURE occupies a different layer: it exposes a callable structured representation whose outputs can be compiled into exact evaluation targets.

## 9 Limitations

The 32-task panel resolves large score differences better than close ones: one answer moves the score by 3.125 points. The English ingredient seeds cover four EPICURE operations, and hosted routes are a dated snapshot. Public prompts may eventually be memorised. The assisted condition names the operation, so it measures execution with a specified tool rather than open-ended tool selection. Finally, agreement with EPICURE inherits the runtime’s representation choices; it is not a food-safety or cultural-authenticity certification. Larger hidden panels, additional operations, and repeated route snapshots are the immediate next tests.

## 10 Conclusion

FLAVOURBENCH turns a versioned culinary runtime into a public automated benchmark. GPT-5.6 Sol Pro leads the 20-model panel with a FlavourBench Score of 62.5. Giving the same endpoints a specified EPICURE operation raises panel accuracy by +50.0 points. The score, gain, completion rates, prompts, responses, and traces all reconstruct from the released 1,280-arm grid. The result is a simple benchmark with a separate, measured intervention: what models know, and what EPICURE adds.



## 232 A Real prompts, answers, and tool calls

233 A benchmark paper should show what was actually asked. The four cases below are selected  
234 mechanically, one per family. Where possible, the builder chooses a model and task that is wrong in  
235 Model only and correct in Model + Epicure. Each case includes the full prompt, both final answers,  
236 the actual operation and arguments, and the exact answer key.

237 **Case 1: Substitution with GPT-5.6 Sol Pro. Prompt.** FlavourBench Epicure-native exact task.  
238 Use the read-only culinary evidence tools if they are available; otherwise answer from your own  
239 knowledge.

240 According to Epicure’s ‘neighbors’ result for ‘tomato’ with top\_k=4, which ingredient is ranked first?

241 Choices: A. oregano B. bell\_pepper C. olive\_oil D. onion

242 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL\_CHOICE:  
243 X’, replacing X with A, B, C, or D.

244 **Model only.** FINAL\_CHOICE: D **Model + Epicure.** FINAL\_CHOICE: B

245 **Epicure call.** neighbors({"ingredient": "tomato", "top\_k": 4}). The answer key is  
246 FINAL\_CHOICE: B.

247 **Epicure result.** {"ingredient": "tomato", "neighbors": [{"name": "bell\_pepper", "rank":  
248 1, "sim": 0.4799}, {"name": "olive\_oil", "rank": 2, "sim": 0.4786}, {"name": "oregano",  
249 "rank": 3, "sim": 0.4646}, {"name": "onion", "rank": 4, "sim": 0.4567}]}

250 **Case 2: Composition with DeepSeek V4 Pro. Prompt.** FlavourBench Epicure-native exact task.  
251 Use the read-only culinary evidence tools if they are available; otherwise answer from your own  
252 knowledge.

253 What exact four-decimal ‘pairing\_score’ does Epicure return for ‘tomato’ and ‘basil’?

254 Choices: A. 0.2707 B. 0.2407 C. 0.3207 D. 0.1907

255 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL\_CHOICE:  
256 X’, replacing X with A, B, C, or D.

257 **Model only.** FINAL\_CHOICE: B **Model + Epicure.** FINAL\_CHOICE: A

258 **Epicure call.** pairing\_score({"ingredient\_a": "tomato", "ingredient\_b":  
259 "basil"}). The answer key is FINAL\_CHOICE: A.

260 **Epicure result.** {"all\_pairs\_range": {"median": 0.092, "p10": 0.0197, "p25": 0.0521, "p75":  
261 0.1392, "p90": 0.1899}, "pairing\_score": 0.2707, "percentile\_label": "very high (>=p90)",  
262 "resolved\_a": "tomato", "resolved\_b": "basil"}

263 **Case 3: Cookability with GPT-5.6 Sol Pro. Prompt.** FlavourBench Epicure-native exact task.  
264 Use the read-only culinary evidence tools if they are available; otherwise answer from your own  
265 knowledge.

266 Which statement matches Epicure’s ‘compare\_on\_axis’ result for ‘coffee’ versus ‘chocolate’ on  
267 ‘cf\_bitter’?

268 Choices: A. coffee has the higher projection B. the projections are equal to four decimals C. Epicure  
269 cannot resolve one or both ingredients D. chocolate has the higher projection

270 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL\_CHOICE:  
271 X’, replacing X with A, B, C, or D.

272 **Model only.** FINAL\_CHOICE: A **Model + Epicure.** FINAL\_CHOICE: D

273 **Epicure call.** compare\_on\_axis({"axis": "cf\_bitter", "ingredient\_a": "coffee",  
274 "ingredient\_b": "chocolate"}). The answer key is FINAL\_CHOICE: D.

275 **Epicure result.** {"axis": "cf\_bitter", "axis\_range": {"p10": -0.0877, "p90": 0.1429},  
276 "delta": 0.0835, "label\_a": "moderate", "label\_b": "high", "projection\_a": 0.0076,  
277 "projection\_b": 0.0911, "resolved\_a": "coffee", "resolved\_b": "chocolate"}

278 **Case 4: Evidence with Grok 4.5. Prompt.** FlavourBench Epicure-native exact task. Use the  
279 read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.  
280 Which cuisine direction has the highest cosine score in Epicure’s ‘cultural\_profile’ result for ‘miso’?  
281 Choices: A. Japanese B. East\_Asian C. Southeast\_Asian D. South\_Asian  
282 In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL\_CHOICE:  
283 X’, replacing X with A, B, C, or D.  
284 **Model only.** {"name": "cultural\_profile", "arguments": {"ingredient":  
285 "miso"}} **Model + Epicure.** FINAL\_CHOICE: A  
286 **Epicure call.** cultural\_profile({"ingredient": "miso"}). The answer key is  
287 FINAL\_CHOICE: A.  
288 **Epicure result.** {"cuisines": {"East\_Asian": {"range": {"p10": -0.1649, "p90":  
289 0.3039}, "score": 0.1625}, "Eastern\_European": {"range": {"p10": -0.1466, "p90":  
290 0.1034}, "score": -0.1494}, "Japanese": {"range": {"p10": -0.1359, "p90": 0.1896},  
291 "score": 0.3227}, "Latin\_American": {"range": {"p10": -0.1672, "p90": 0.1281},  
292 "score": -0.1538}, "Mediterranean": {"range": {"p10": -0.2323, "p90": 0.206}, "score":  
293 -0.1198}, "South\_Asian": {"range": {"p10": -0.0808, "p90": 0.071}, "score": -0.0585},  
294 "Southeast\_Asian": {"range": {"p10": -0.1625, "p90": 0.193}, "score": 0.0426},  
295 "Western\_Atlantic": {"range": {"p10": -0.2304, "p90": 0.183}, "score": -0.1546}},  
296 "note": "Cosine similarity to each cuisine direction. Use p10/p90 to judge whether a score  
297 is typical or extreme.", "resolved": "miso"}  
298 These examples show exactly what the two measurements contain. Model only tests knowledge or  
299 inference about the hidden graph. Model + Epicure names the operation, so it tests whether the model  
300 can form valid arguments, read the returned structure, and preserve the answer contract. A correct  
301 choice with the wrong operation still earns answer credit, but the release separately reports whether  
302 the reference operation was matched. Across the run there are 606 projected EPICURE calls and 606  
303 model and task pairs matching the reference operation.

## 304 **B Additional result figures**

## 305 **C Exact execution routes**

## 306 **D Protocol details**

307 All arms request temperature 0, top- $p$  1, and seed 20260810 where the provider supports it. The final  
308 response budget is 2,048 tokens. Model + Epicure has one call round, one call total, a 4,096-byte  
309 result bound, and a 4,096-byte cumulative result bound. At most two provider attempts are allowed,  
310 and an ambiguously delivered request is never replayed. The final marker is parsed from plain text for  
311 transport portability; direct providers may use native structured generation internally and normalize  
312 to the same marker.

## 313 **E Data fields**

314 Each observation binds model ID, task ID, condition, source status, source and response hashes,  
315 returned model and provider, finish reason, response time, final answer, observed and expected choice,  
316 normal-parse flag, correctness, and a projected Epicure trace. Each trace event contains its name,  
317 structured arguments, error flag, response time, and result hash. The release does not require raw  
318 provider credentials or hidden service state.

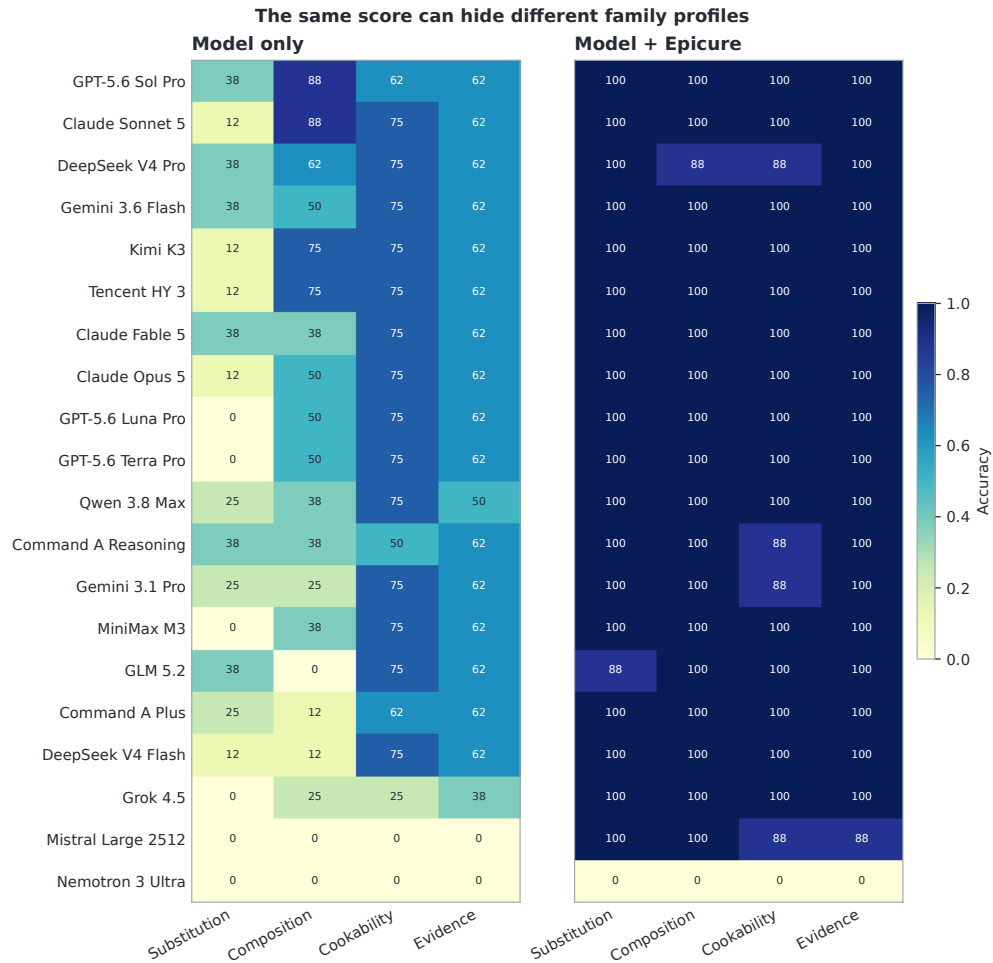


Figure 4: Model only and Model + Epicure accuracy by task family. Each value is a percentage over eight tasks. Similar FlavourBench Scores can come from different family profiles.

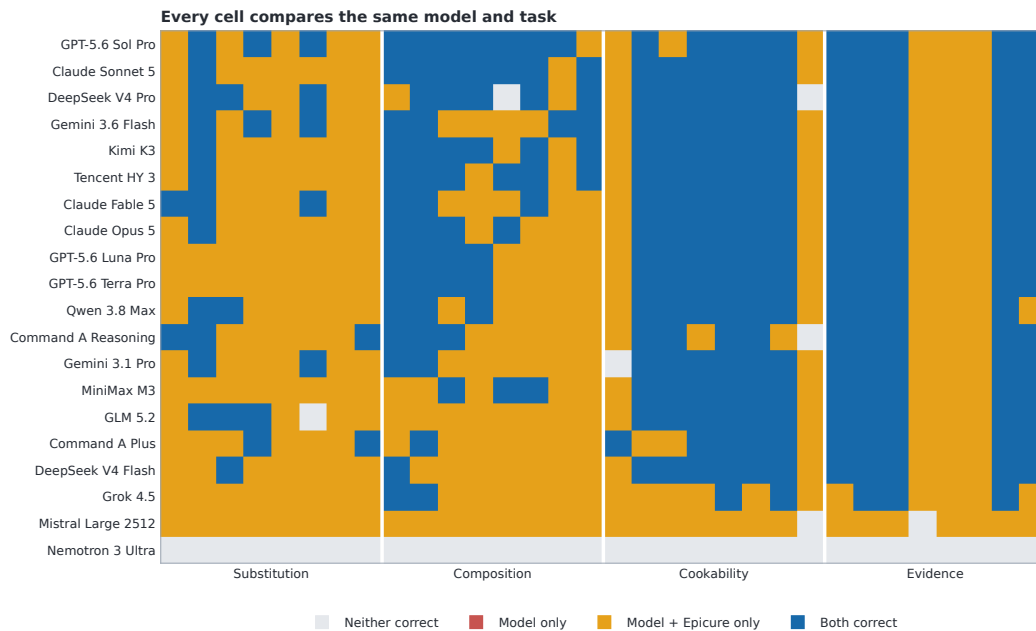


Figure 5: All 640 matched outcomes, ordered by FlavourBench Score and grouped by task family. The matrix shows which exact tasks produce Epicure Gain and where the assisted response regresses.

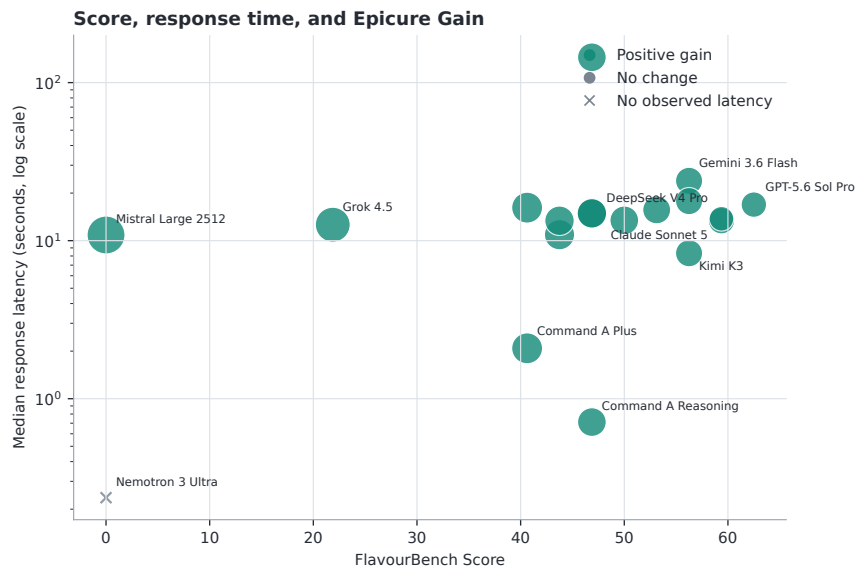


Figure 6: FlavourBench Score, response time, and Epicure Gain. Response time uses a log scale. Point size increases with the magnitude of gain; colour indicates its direction. An  $\times$  marks a route with no observed response time.

| Slot | Model               | Execution route                          |
|------|---------------------|------------------------------------------|
| 1    | GPT-5.6 Sol Pro     | openrouter: openai/flex                  |
| 6    | Claude Sonnet 5     | openrouter: amazon-bedrock/claude-on-aws |
| 13   | DeepSeek V4 Pro     | openrouter: baidu/fp8                    |
| 8    | Gemini 3.6 Flash    | openrouter: google-vertex/global/flex    |
| 10   | Kimi K3             | kimi_direct: kimi-code-direct            |
| 18   | Tencent HY 3        | openrouter: baidu/fp8                    |
| 4    | Claude Fable 5      | openrouter: amazon-bedrock               |
| 5    | Claude Opus 5       | openrouter: amazon-bedrock/claude-on-aws |
| 3    | GPT-5.6 Luna Pro    | openrouter: openai/flex                  |
| 2    | GPT-5.6 Terra Pro   | openrouter: openai/flex                  |
| 11   | Qwen 3.8 Max        | openrouter: alibaba                      |
| 20   | Command A Reasoning | cohere_direct: cohere-direct             |
| 7    | Gemini 3.1 Pro      | openrouter: google-vertex/global/flex    |
| 15   | MiniMax M3          | openrouter: together                     |
| 12   | GLM 5.2             | openrouter: coreweave/fp4                |
| 19   | Command A Plus      | cohere_direct: cohere-direct             |
| 14   | DeepSeek V4 Flash   | openrouter: cloudflare/fp8               |
| 9    | Grok 4.5            | openrouter: xai/zdr                      |
| 17   | Mistral Large 2512  | openrouter: mistral                      |
| 16   | Nemotron 3 Ultra    | openrouter: deepinfra/fp4                |

Table 5: All 20 model routes in the released leaderboard. Routes are selected before generation; automatic fallback is disabled.

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